# Privacy and data governance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

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WORKBOOK · 1 OF 15

## Recall and choose your starting point

> Record your starting explanation for Privacy and data governance before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Research ethics and consent

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Sensitivity analysis

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Statistical tests and assumptions

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 15

## Practise with fading help

> Complete the supported Privacy and data governance trace and record the first step you can explain without help.

Use W2: Trace the worked case for Privacy and **data governance**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

## Individual reasoning check

> Check the proposed Privacy and data governance explanation against the supplied facts before discussing it.

Use the worked case for Privacy and **data governance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 15

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Apply a new case independently

Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than5. Decide.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 15

## W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Privacy and data governance facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

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WORKBOOK · 6 OF 15

## W2 Trace the worked case

> Trace the worked case. Apply the supplied Privacy and data governance facts and keep your reasoning.

Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

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## W3 Audit the limit

> Audit the limit. Apply the supplied Privacy and data governance facts and keep your reasoning.

A report omits this qualification: A size threshold or coding scheme alone does not guarantee privacy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

WORKBOOK · 8 OF 15

## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Privacy and data governance facts and keep your reasoning.

Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than5. Decide.

**My reasoning and evidence**

WORKBOOK · 9 OF 15

## W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Privacy and data governance facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

WORKBOOK · 10 OF 15

## Project checkpoint and reflection

> Connect Privacy and data governance to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled privacy and **data governance evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of privacy and **data governance**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

**My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 12 OF 15**

## Pictorial case practice: Privacy and data governance

> Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names. Explain the labelled step before checking the changed condition.

Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.

The analysis table holds codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling.

AI ACADEMY / AI-7/20

# Privacy and data governance

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Separate tables

Keep analysis records apart from the identity key.

### 2. Limit access

Analysts receive scores and codes, not the name linkage.

### 3. Define retention

Apply the approved deletion schedule to both stores and listed copies.

**CHANGE ONE THING**

Publish the identity key with the analysis table.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Point to the analysis table and the restricted identity key. Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 14 OF 15

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Apply the approved deletion schedule to both stores and listed copies.

1. Define retention

2. Separate tables

3. Limit access

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Define retention Apply the approved deletion schedule to both stores and listed copies.

**Pseudonym**ous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

### Practice 2: explain the deciding evidence

#### Changed case · predict first

Changed-case practice: Publish the identity key with the analysis table. Which conclusion is supported by the supplied facts?

1. The records become directly linkable to names and breach this access design.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

#### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The records become directly linkable to names and breach this access design. Separation only helps while the linkage key remains restricted.

### Repair a missing building block

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

WORKBOOK · 15 OF 15

# Visual concept journey

> Explain each Privacy and data governance picture from its labels, then record a relevant limit.

### 01. Privacy and data governance: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

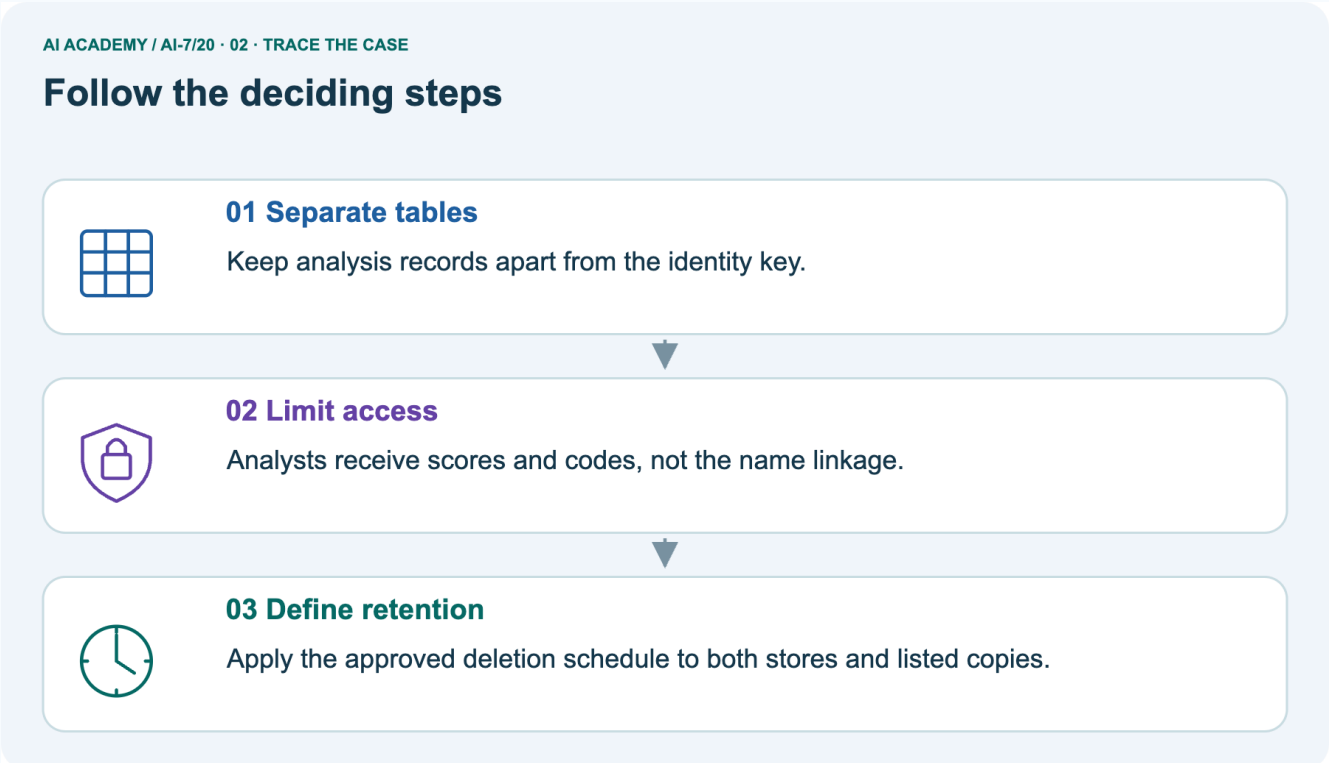
Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names. Supplied facts: The analysis table holds codes and scores.

A separate restricted key links codes to names only for authorised withdrawal handling.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



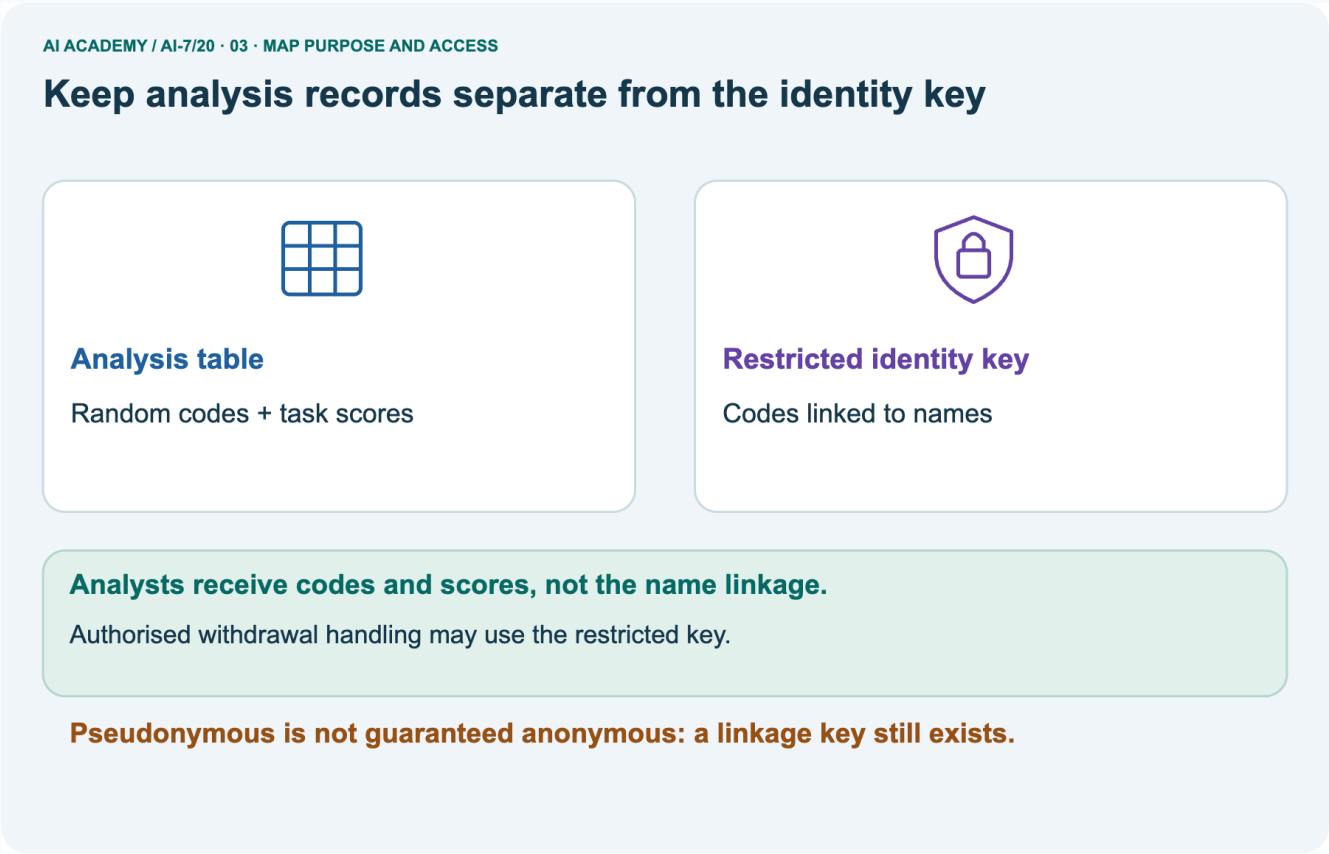

Follow the labelled picture with the matching explanation. Open the vector diagram

Separate tables: Keep analysis records apart from the identity key. → Limit access: Analysts receive scores and codes, not the name linkage. → Define retention: Apply the approved deletion schedule to both stores and listed copies.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Keep analysis records separate from the identity key



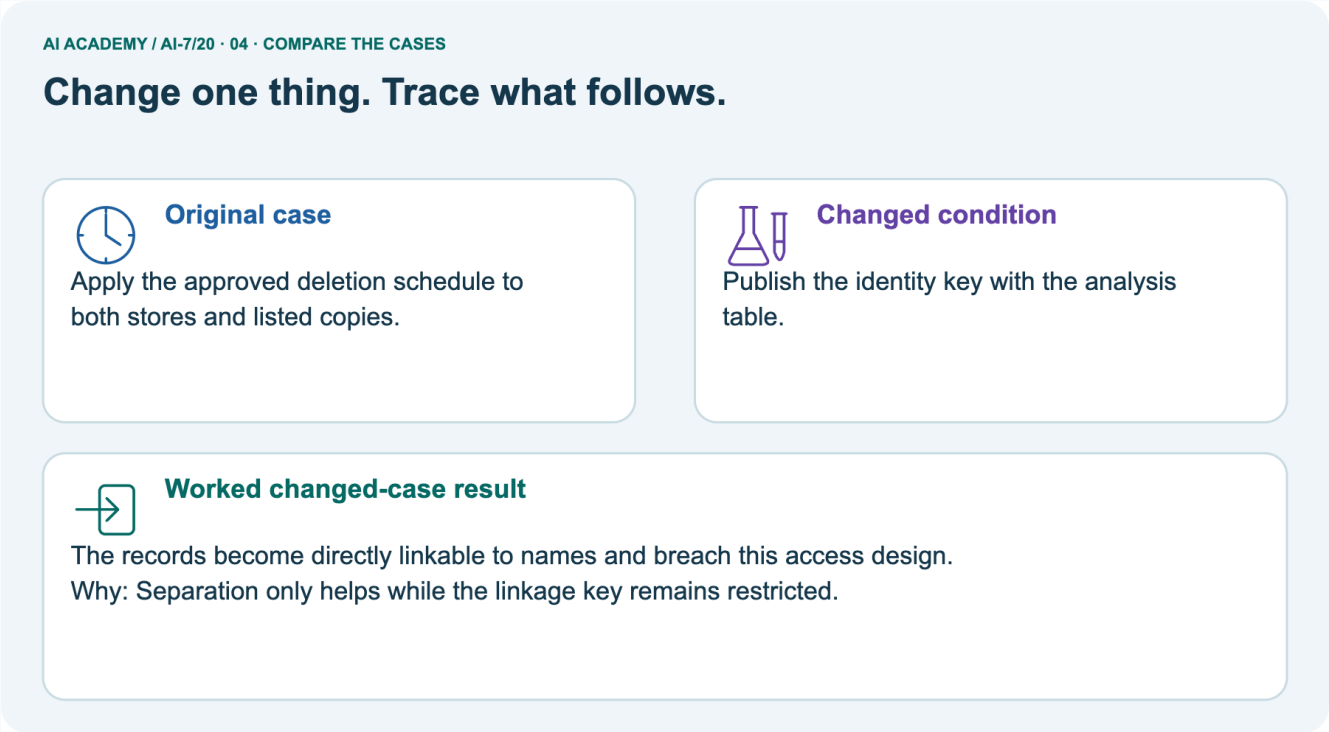

Follow the labelled picture with the matching explanation. Open the vector diagram

Analysts receive random participant codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling. Keeping the key restricted reduces exposure; it does not guarantee anonymity.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

**04. Change one thing. Trace what follows.**




Follow the labelled picture with the matching explanation. Open the vector diagram

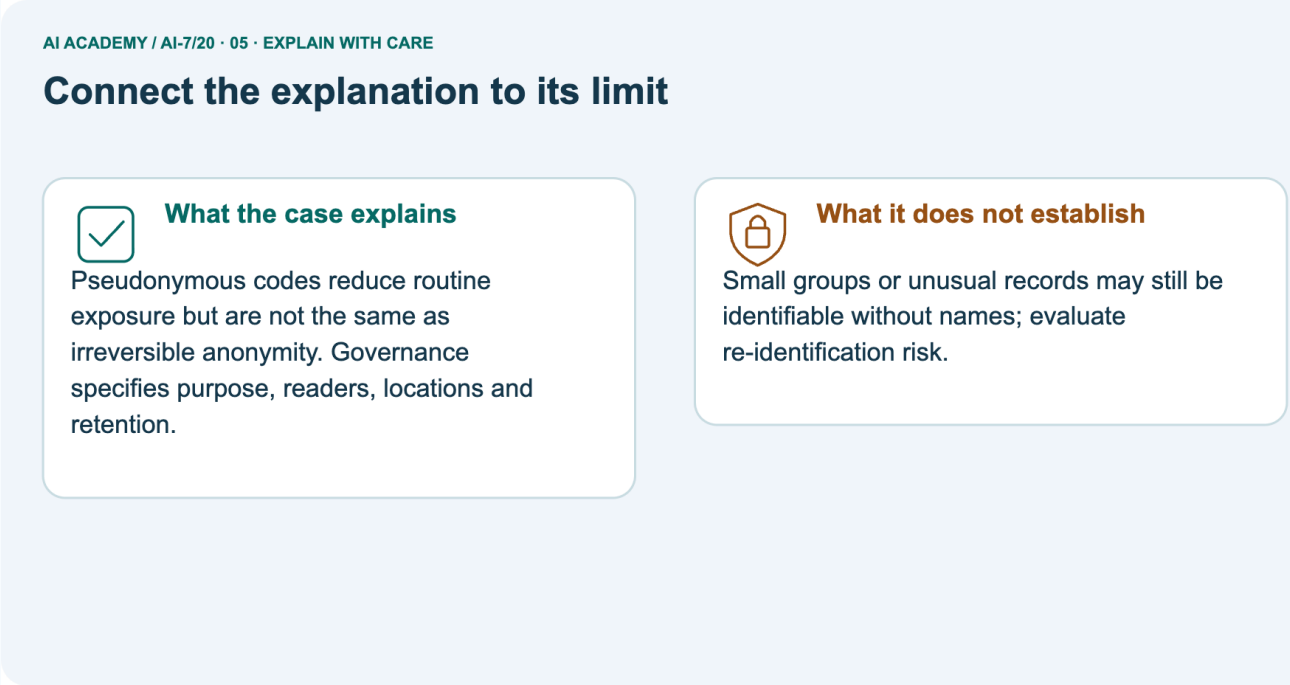
Original case: Apply the approved deletion schedule to both stores and listed copies. Changed condition: Publish the identity key with the analysis table.

Expected result: The records become directly linkable to names and breach this access design. Reason: Separation only helps while the linkage key remains restricted.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

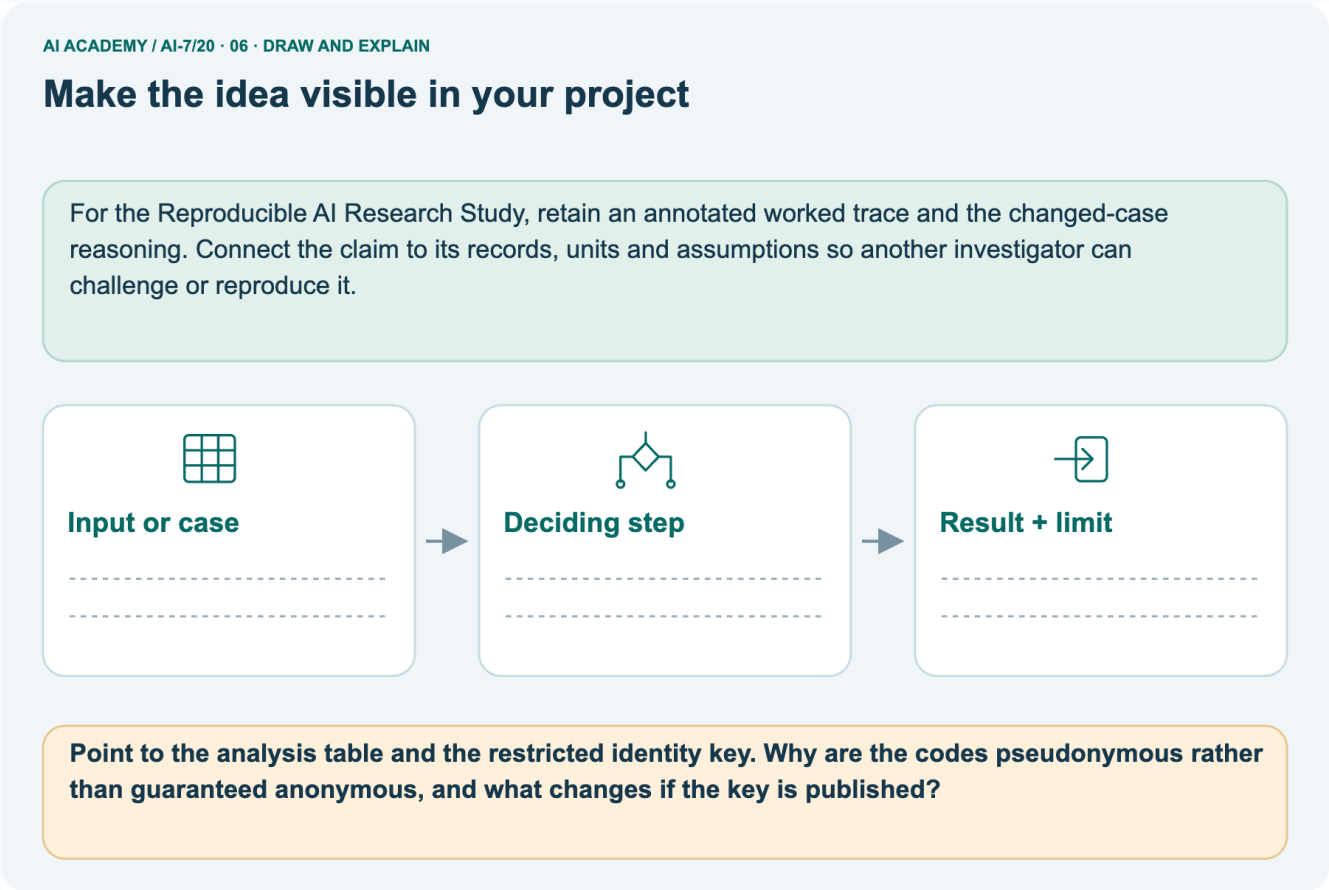
**Pseudonym**ous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

Limit: Small groups or unusual records may still be identifiable without names; evaluate re-identification risk.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the analysis table and the restricted identity key.

Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

> Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.